#1.

The M.C is irreducible, then every state communicates with others. Thus $P_{ij} = P(X_n = j \text{ for some } n \geq 0 \mid X_0 = i) > 0$

$i \neq j$ The MC is recurrent. then the M.C will visit j for finite steps. starting at j.

then i can definitely visit any j in finite steps with probabilities 1.

2.

(a) $P(X_n = j \mid X_{n-1} = i, X_{n-2} = k, \cdots X_0 = p)$

$= P(X_n = j \mid X_{n-1} = i)$

$\therefore$ set $\{X_n, n \geq 0\}$ is a DTMC

$P_{ii} \geq P_{i,i+1} \cdot P_{i+1,i} > 0$

P_{ij}. if $j > i$. $P_{ij} \geq P_{i,i+1} P_{i+1,i+2} P \cdots P_{j-1,j} > 0$

if $j < i$ $P_{ij} \geq P_{i,i-1} P_{i-1,i-2} P \cdots P_{j+1,j} > 0$

$\therefore P_{ij} > 0$ for any $i, j \geq 0$

$\therefore \{X_n, n \geq 0\}$ is a irreducible DTMC

For transition probability matrix, $P' = (P'_{i,j})$. $i, j \geq 0$

$P'_{i,i+1} = \lambda(1-\mu)$ $P'_{i,i} = \lambda\mu + (1-\lambda)(1-\mu)$

$P'_{i+1,i} = (1-\lambda)\mu$ for $i \geq 1$ $\quad P_{00} = (1-\lambda)$ $P_{01} = \lambda$

others $P_{i+1,j} = 0$ $\quad j \neq i, i+1, i+2$, $i \geq 0$

state i : $X_n = i$ state j : $X_{n+1} = j$

(b) $Y_{i+1} = Y_i + 1$ or $Y_{i+1} = Y_i - 1$

$\therefore$ Y_{i+1} is only related to Y_i

$\therefore \{Y_m, m \geq 0\}$ is a DTMC

$\text{Prob}(Y_{m+1} = Y_m + 1) = \lambda$ $\quad \text{Prob}(Y_{m+1} = Y_m - 1) = \mu$

$\therefore \text{Prob}(Y_n = j \text{ for some } n \geq 0 \mid Y_0 = i) > 0$, $i, j \geq 0$

$\therefore \{Y_m, m \geq 0\}$ is a irreducible DTMC

For transition probability matrix

$P' = (P'_{ij})$ state i : $Y_m = i$ state j : $Y_{m+1} = j$

$$P_{ij} = \begin{cases} 0, & i = j \\ \frac{\lambda(1-\mu)}{1-\lambda\mu-(1-\lambda)(1-\mu)} & j = i+1 \\ \frac{\mu(1-\lambda)}{1-\lambda\mu-(1-\lambda)(1-\mu)} & i = j+1 \\ 0, & \text{o.w.} \end{cases} \quad , i, j \geq 1$$

$P_{01} = 1$

$P_{0j} = 0$, $j = 0$ or $j > 1$

(c) if $\mu > \lambda$, the MC is transient

$\mu(1-\lambda) > \lambda(1-\mu)$

if $\mu \leq \lambda$, the MC is recurrent

$\mu(1-\lambda) < \lambda(1-\mu)$

(d) $(1-\lambda)\mu > \lambda(1-\mu)$

$\therefore \mu > \lambda$ is transient, because for state 1, will alway having a bigger probability to state 0

(e)

#5

(a) X_n = # of label after nth operation

$P_{i,(i+1) \bmod (n+1)} = q_i p \quad P_{i,(i-1) \bmod (n+1)} = q_i (1-p)$

$P_{i,i} = 1 - q_i$

if $X_1 = 1$, $P(X_1 = 1 \mid X_0 = 0) = q_0 p$

set it as a gambler ruin game, and the absorbing value is 0 and i

$P_i = 1 \quad P_0 = 0$

$$P_k = P(X_n=k \mid X_{n-1}=k-1)P_{k-1} + P(X_n=k \mid X_{n-1}=k)P_k + P(X_n=k \mid X_{n-1}=k+1)P_{k+1}$$

$$= q_{k-1}P\,P_{k-1} + (1-q_k)P_k + q_{k+1}(1-P)P_{k+1}$$

$$q_{k+1}P_{k+1} - q_kP_k = \frac{P}{1-P}(q_kP_k - q_{k-1}P_{k-1})$$

$$\vdots$$

$$q_2P_2 - q_1P_1 = \frac{P}{1-P}q_1P_1$$

$$q_iP_i - q_{i-1}P_{i-1} = \frac{P}{1-P}(q_{i-1}P_{i-1} - q_{i-2}P_{i-2}) = \left(\frac{P}{1-P}\right)^{i-1}q_1P_1$$

$$\therefore q_iP_i = \left[\left(\frac{P}{1-P}\right)^{i-1} + \left(\frac{P}{1-P}\right)^{i-2} + \cdots + \left(\frac{P}{1-P}\right)^1 + 1\right]q_1P_1$$

$$= \begin{cases} \dfrac{\left(\frac{P}{1-P}\right)^i - 1}{\frac{P}{1-P} - 1}\,q_1P_1 & \text{if } P \neq \frac{1}{2} \\ i\,q_1P_1 & \text{if } P = \frac{1}{2} \end{cases}$$

$$\therefore P_1 = \begin{cases} \dfrac{\frac{P}{1-P} - 1}{\left(\frac{P}{1-P}\right)^i - 1}\,\dfrac{1}{q_iq_1} & \text{if } P \neq 1/2 \\ \dfrac{q_i}{iq_1} & \text{if } P = 1/2 \end{cases}$$

if $X_1 = n$ $\quad P(X_1 = n \mid X_0 = 0) = q_i(1-P)$

change "i" to "n", "P" to "$1-P$"

$$P_n = \begin{cases} \dfrac{\frac{1-P}{P} - 1}{\left(\frac{1-P}{P}\right)^{n-i+1} - 1}\,\dfrac{1}{q_iq_n} & \text{if } P \neq 1/2 \\ \dfrac{q_i}{iq_n} & \text{if } P = 1/2 \end{cases}$$

$$\therefore F_i = \left(P_1 P(X_1=1 \mid X_0=0) + P_n P(X_1=n \mid X_0=0)\right) \Big/ P(X_1=1 \text{ or } X_1=n \mid X_0=0)$$

$$= \begin{cases} \dfrac{\frac{p}{1-p}-1}{(\frac{p}{1-p})^i-1}\dfrac{p}{q_i q_1} + \dfrac{\frac{1-p}{p}-1}{(\frac{1-p}{p})^{n-i+1}-1}\dfrac{1-p}{q_i q_n} & \text{if } p \neq 1/2 \\ \dfrac{q_i}{2i\,q_1} + \dfrac{q_i}{2i\,q_n} & \text{if } p = 1/2 \end{cases}$$

(b) $$P = \left[P_1(i=n)P(X_1=n \mid X_0=0) + P_n(i=1)P(X_1=1 \mid X_0=0)\right] \Big/ P(X_1=1 \text{ or } n \mid X_0=0)$$

$$= \begin{cases} \dfrac{\frac{p}{1-p}-1}{(\frac{p}{1-p})^n-1}\dfrac{p}{q_n q_1} + \dfrac{\frac{1-p}{p}-1}{(\frac{1-p}{p})^n-1}\dfrac{1-p}{q_1 q_n}, & \text{if } p \neq 1/2 \\ \dfrac{q_n}{2n\,q_1} + \dfrac{q_1}{2\,q_n} & \text{if } p = 1/2 \end{cases}$$

(c) Set A: robot inspects all machine at least once **before** go back to 0 for the first time

$$P = P\left((X_1=1 \mid X_0=0) \,\middle|\, A\right)$$

$$= \begin{cases} \dfrac{(\frac{p}{1-p}-1)p}{(\frac{p}{1-p})^n-1} \Big/ \left(\dfrac{(\frac{p}{1-p}-1)p}{(\frac{p}{1-p})^n-1} + 1-p\right), & \text{if } p \neq 1/2 \\ \dfrac{q_n}{n\,q_1} \Big/ \left(\dfrac{q_n}{n\,q_1} + \dfrac{q_1}{q_n}\right), & \text{if } p = 1/2 \end{cases}$$

#4